A3-PS001-TC001

FEDERATED Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem Description

Centralized traffic forecasting

Traffic data from multiple intersections normally needs to be transferred to one central server, creating continuous communication overhead.

HOW ARE MODEL ADDRESSES: Each simulated intersections train locally, and the server receives model updates.

Privacy and data movement

Raw traffic observations from different locations increases data exposure and data transfer requirements.

HOW ARE MODEL ADDRESSES: The federated workflow keeps raw training samples in their assigned client partitions.

Non-IID traffic patterns

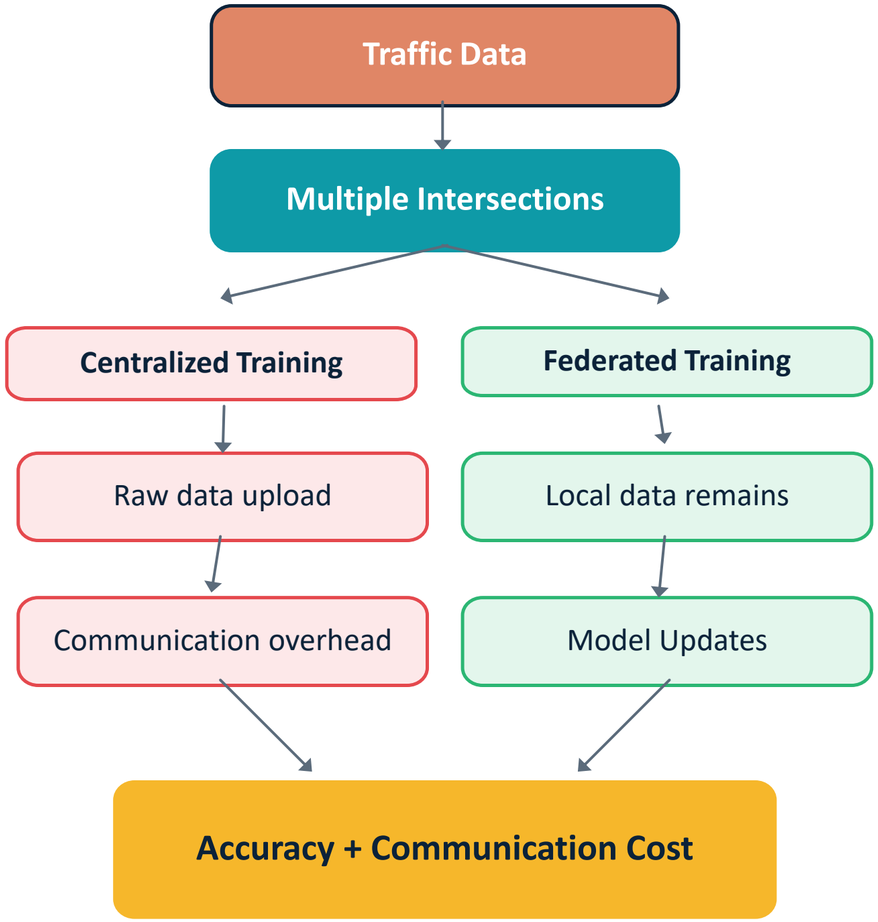
Different intersections have different vehicle counts, different traffic distributions, making federated training more challenging.

HOW ARE MODEL ADDRESSES: it compares IID and non-IID training to investigate how different data distributions affect prediction accuracy.

Communication overhead

Every federated training round requires model parameters to be sent to clients and updated parameters to be returned to the server. This consumes bandwidth.

HOW ARE MODEL ADDRESSES: it measures upload bytes,download bytes,cumulativr communication cost, and convergence.

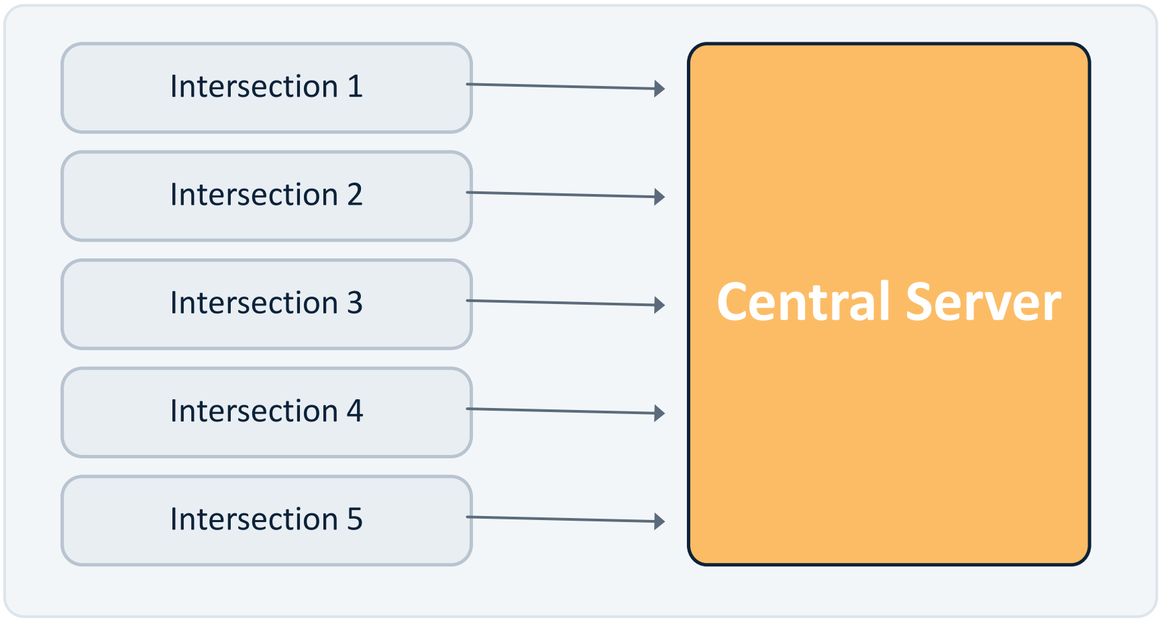


WHAT ARE MODEL IS DOING : how can Federated Learning predict future traffic counts using traffic dataset while evaluating prediction accuracy,convergence and communication cost?

Background

TRADITIONAL APPROACH

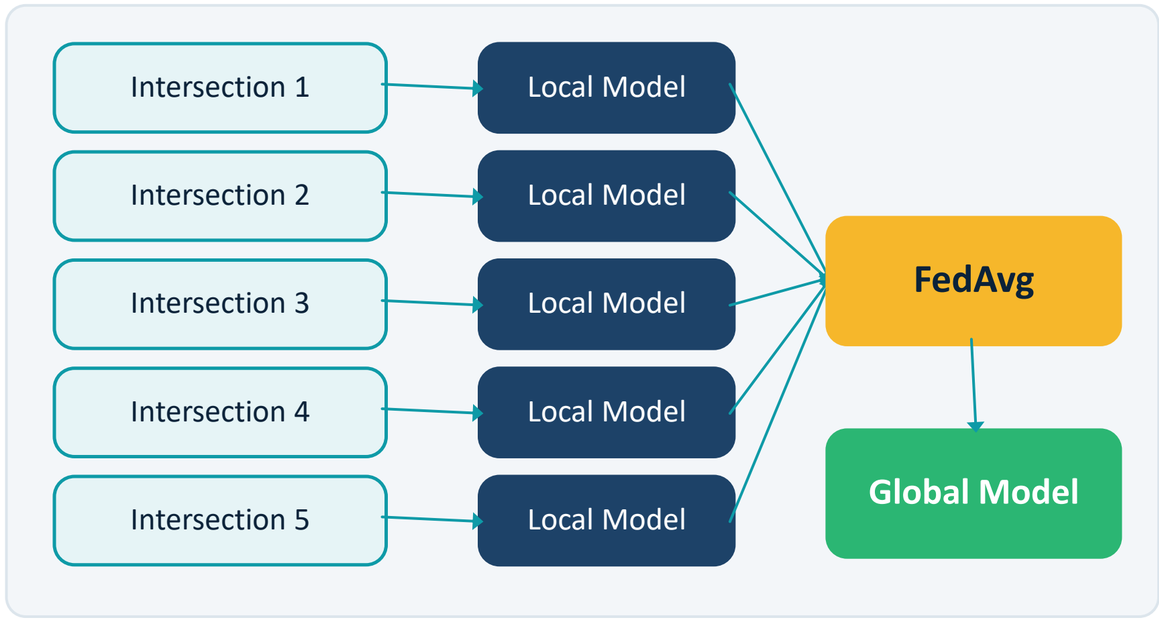
Centralized Learning



Raw traffic data is sent to a central location.

PROPOSED CONCEPT

Federated Learning



Key Idea: Keep traffic data local and communicate model updates instead of raw observations.

"Our project simulates intersections using a public traffic dataset rather than deploying physical cameras or roadside sensors."

Literature & RFC Review

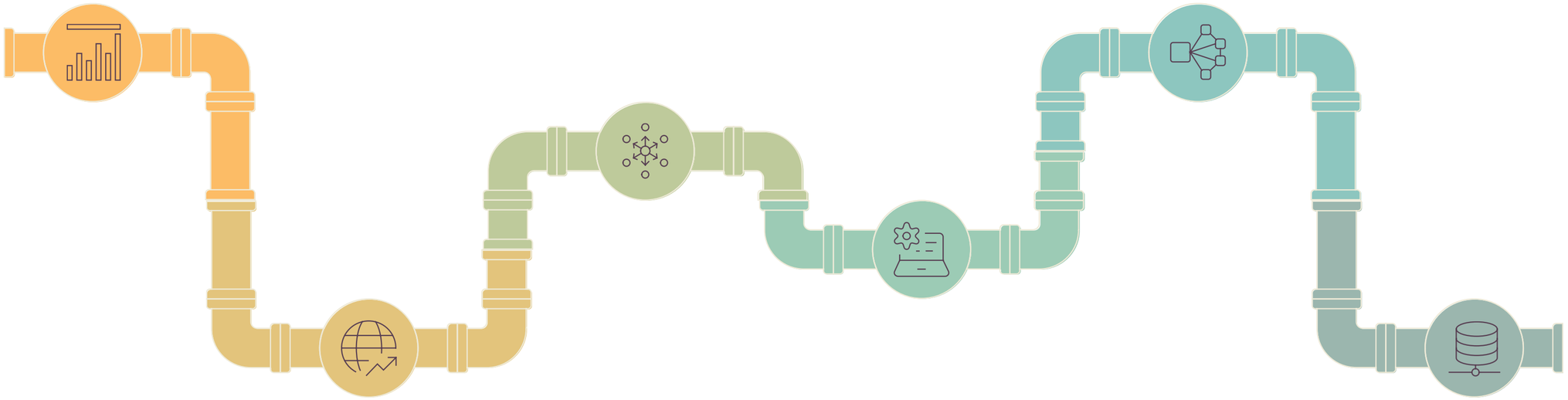
Work	Contribution	Limitation / Gap
McMahan et al. — FedAvg	Federated averaging enables decentralized model training without centralizing raw data	Communication and client heterogeneity remain important
Centralized Traffic Forecasting	Uses aggregated traffic data for strong prediction performance	Requires centralized data transfer
Federated Traffic Forecasting	Keeps training data distributed across simulated clients	Non-IID data can affect convergence and accuracy

Research Gap

Existing federated learning approaches demonstrate decentralized training, but our model focuses specifically on traffic-flow forecasting and communication cost together under stimulated IID and non-IID intersection data.



Proposed Solution



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Evaluation

Compare centralized, local-only, persistence, and federated approaches.

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Federated Aggregation

The server combines client model parameters using FedAvg

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local Training

Each client trains a GRU to learn patterns in vehicle count.

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Client Stimulation

Create five stimulated clients and evaluate IID and non-IID training partition.

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Preprocessing

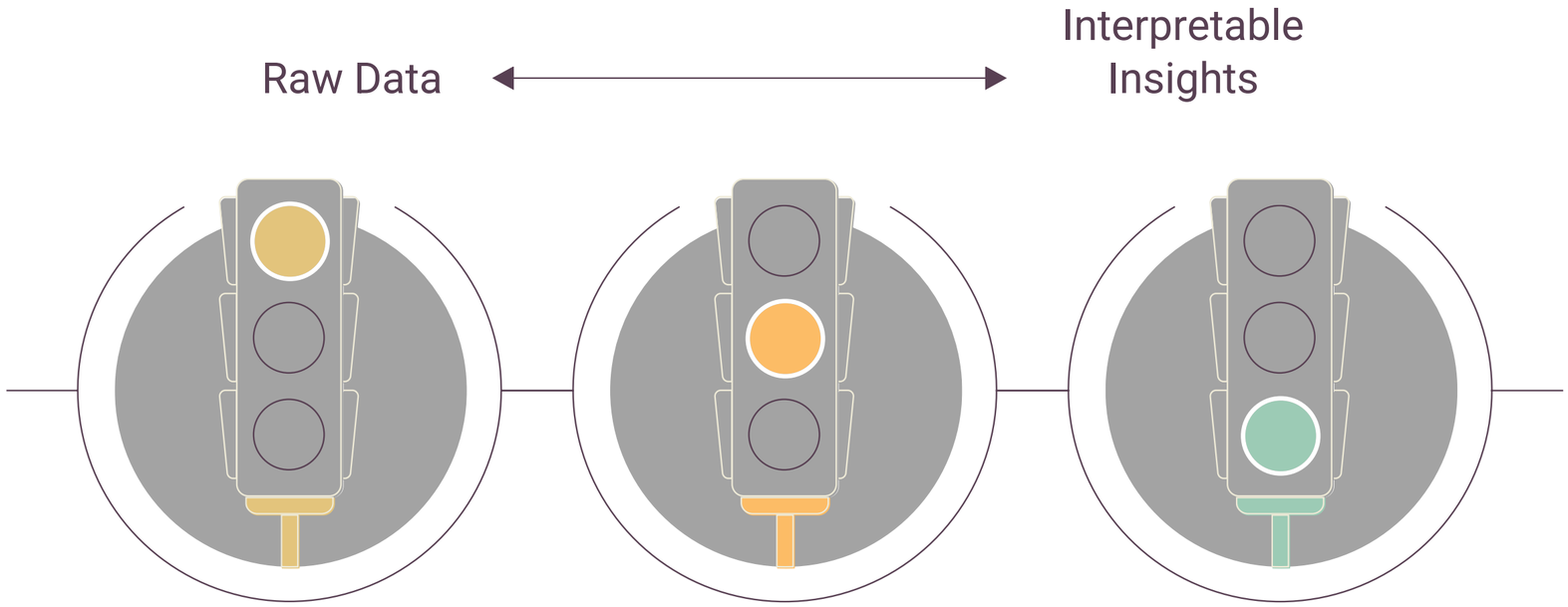
Partition the dataset into 5 simulated intersections.

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Dataset Selection

Use a public traffic-flow dataset

Solution Optimisation



Local Training

Processes data locally, enhancing privacy

Round-wise Convergence

Monitors progress and ensures reliability

Human-Readable Explanations

Converts results into clear, actionable insights

Accuracy

- Evaluate MAE and RMSE on a held-out chronological test set.
- Compare centralized, local-only,
- persistence, and federated approaches.

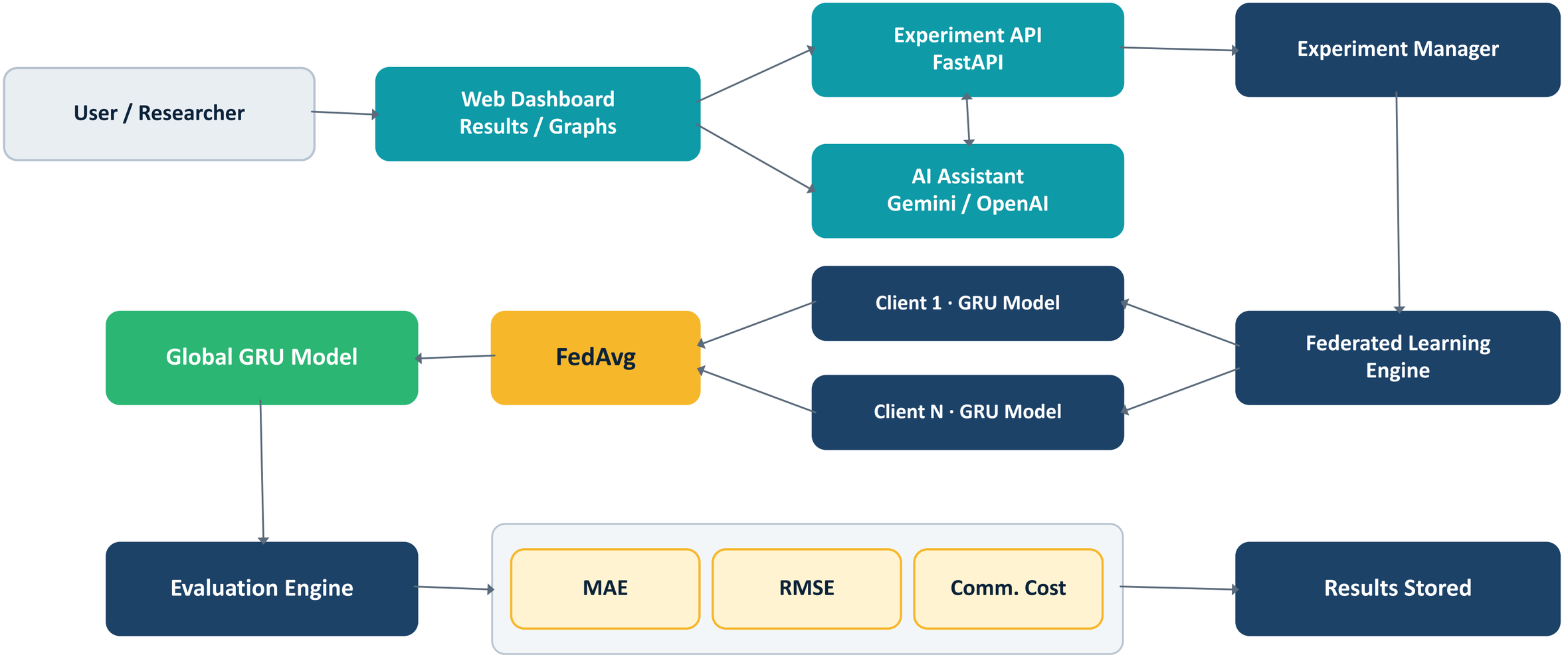
Client Heterogeneity

- Run IID and non-IID experiments.
- Compare how data distribution affects prediction and convergence.

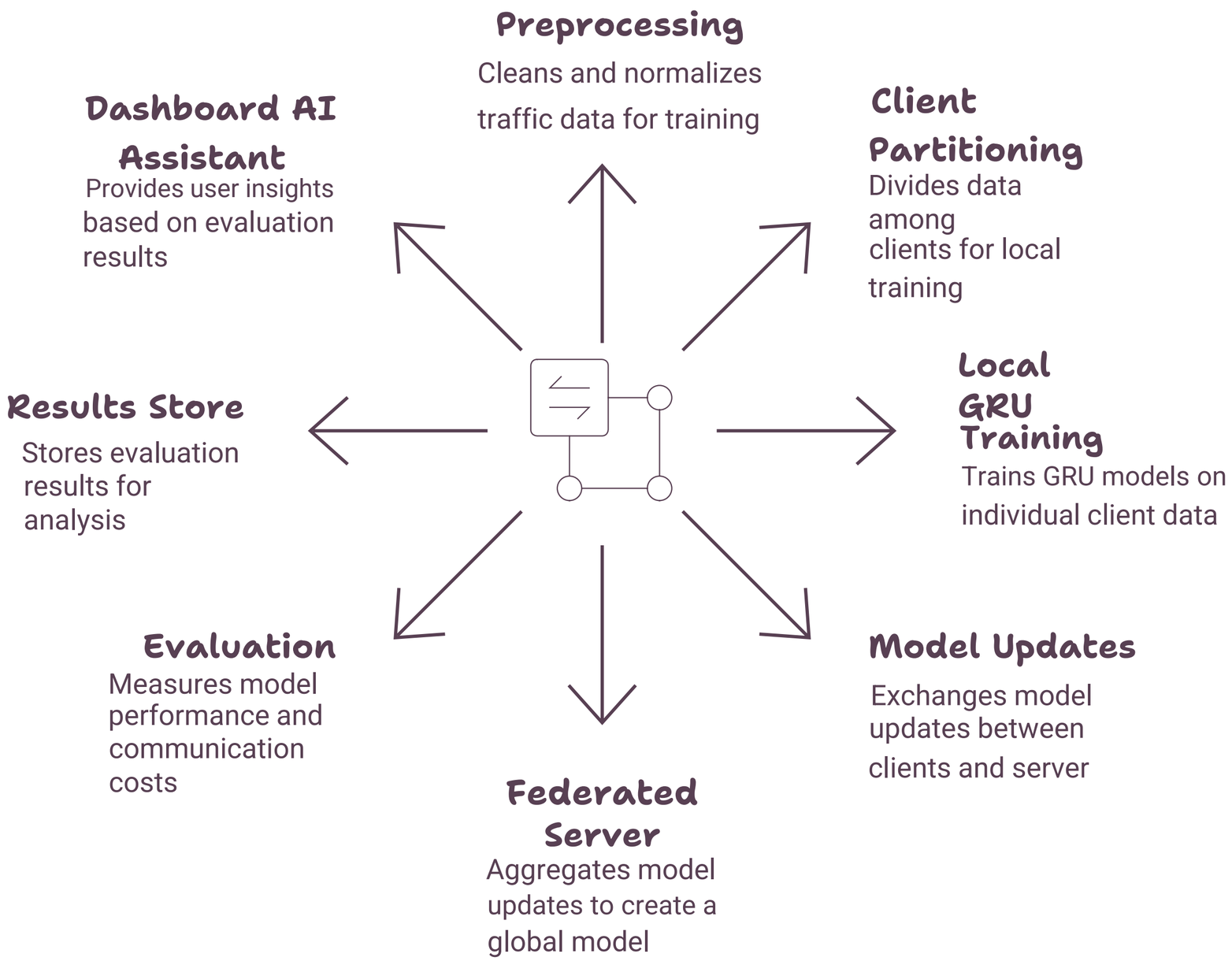
Communication Efficiency

- Record model payload bytes sent from server to clients.
- Record model update bytes returned by clients.
- Calculate total and cumulative communication per round.

Solution Architecture

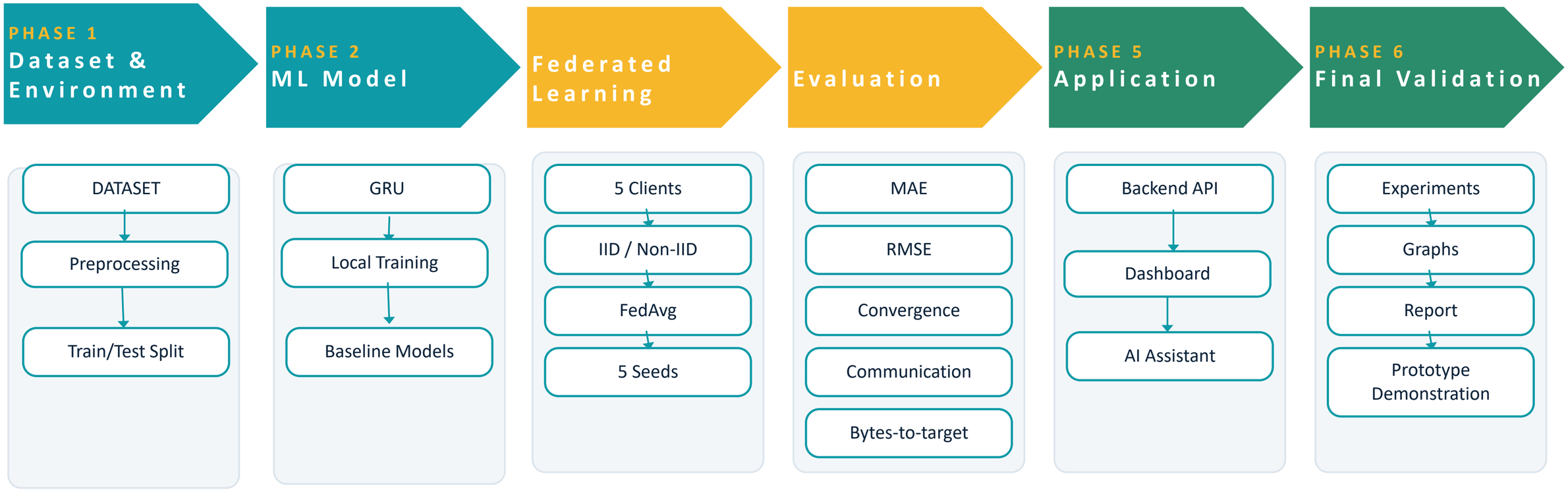


Data Flow Diagram



Raw traffic data does not travel between clients and server; only model updates are exchanged.

Implementation & Deployment Timeline



OVERALL FLOW



Bibliography

01

IEEE Xplore Digital Library. Research papers on federated learning and traffic-flow prediction. https://ieeexplore.ieee.org/

02

ResearchGate. Research publications on federated learning and traffic forecasting. https://www.researchgate.net/

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Association for Computing Machinery (ACM). ACM Digital Library: Research on machine learning and intelligent transportation systems. https://dl.acm.org/

04

Google Scholar. Scholarly literature on federated learning, traffic prediction, and communication efficiency. https://scholar.google.com/

THANK YOU

